

Exact computation and cubic-order growth of the expanded derivatives of x^{x^x}

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Abstract

Let $a(n)$ be the number of nonzero monomials in the fully expanded n th derivative of x^{x^x} , after collecting identical terms. We replace repeated symbolic differentiation by a seven-transition integer recurrence on three exponents. It computes the prefix through n in $O(n^4)$ integer arithmetic operations and $O(n^3)$ live integers. A faster implementation performs the recurrence modulo a machine-size integer and verifies every possible zero by an independent exact coefficient formula; this method is deterministic, not a probabilistic zero test. We prove the explicit bounds

$$\frac{7n^3 + 39n^2 + 26n + 3\varepsilon_n(n-1)}{24} \leq a(n) \leq \frac{2n^3 + 3n^2 + 4n}{3}, \quad n \geq 1,$$

where ε_n is 1 for odd n and 0 for even n . In particular, $a(n) = \Theta(n^3)$. The upper bound counts an explicit lattice envelope; the lower bound uses nonvanishing leading coefficients and high-multiplicity roots of coefficient polynomials. We also identify an unavoidable family of logarithm-free cancellations and an infinite family with positive logarithm exponent. The stronger statement $a(n) \sim \frac{2}{3}n^3$, and the still stronger linear-deficit conjecture, are *not proved here*. Exact code, tests, terms through $n = 100$, and selected computations through $n = 200$ accompany this article.

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1 The problem and the scope of the results

The sequence A281434 is defined in the OEIS as the number of terms in the fully expanded n th derivative of x^{x^x} , with offset 0 [1]. Exponentiation is right-associated. Thus the function in question is

$$f(x) = x^{(x^x)} = \exp(x^x \log x), \quad x > 0,$$

not $(x^x)^x = x^{x^2}$. Its initial term counts are

$$1, 3, 12, 30, 64, 113, 188, 285, 415, 577, 780, 1017, 1312, \dots$$

The positivity restriction supplies an unambiguous real logarithm; all subsequent coefficient manipulations are exact polynomial identities.

The phrase “fully expanded” must include collecting like terms and deleting zero coefficients. Counting product-rule branches is a different problem. Even recording every monomial that could be reached by differentiation gives an overcount: distinct contributions can cancel exactly.

For $n \geq 1$, define

$$B(n) = \frac{2n^3 + 3n^2 + 4n}{3}, \quad \Delta(n) = B(n) - a(n). \quad (1.1)$$

The polynomial $B(n)$ is A037237($n - 1$) [3]. The related OEIS entry A352697 records the differences $\Delta(n)$ and conjectures both $\Delta(n) = O(n)$ and infinitely many zero differences [2]. These observations are useful motivation, but neither is needed by our algorithms or proofs.

Theorem 1.1 (Main results). *There is a deterministic exact algorithm for A281434 requiring $O(n^4)$ integer arithmetic operations and $O(n^3)$ stored integers to compute $a(0), \dots, a(n)$. There is also a machine-modular implementation for a single $a(n)$, with exact verification of every candidate cancellation. For $n \geq 1$,*

$$L(n) := \frac{7n^3 + 39n^2 + 26n + 3\varepsilon_n(n-1)}{24} \leq a(n) \leq B(n), \quad \varepsilon_n = \frac{1 - (-1)^n}{2}. \quad (1.2)$$

Consequently

$$a(n) = \Theta(n^3), \quad \lim_{n \rightarrow \infty} \frac{\log a(n)}{\log n} = 3. \quad (1.3)$$

The distinction between a growth *order* and an asymptotic *equivalent* is important. Equation (1.3) is unconditional. An equivalent $a(n) \sim \frac{2}{3}n^3$ would additionally require $\Delta(n) = o(n^3)$; the present bounds do not establish that. The computations are consistent with a much smaller deficit, but this numerical observation will not be substituted for a nonvanishing theorem.

2 A canonical differential-polynomial representation

2.1 Three auxiliary variables

Put

$$u = x^x, \quad v = x^{-1}, \quad z = \log x, \quad A = z(z + 1).$$

The three elementary derivatives and the logarithmic derivative of f are

$$u' = u(z + 1), \quad v' = -v^2, \quad z' = v, \quad \frac{f'}{f} = u(A + v). \quad (2.1)$$

There is a unique polynomial $P_n \in \mathbb{Z}[u, v, z]$ such that

$$f^{(n)}(x) = f(x)P_n(x^x, x^{-1}, \log x). \quad (2.2)$$

Write

$$P_n(u, v, z) = \sum_{k, j, \ell \geq 0} c_n(k, j, \ell) u^k v^j z^\ell. \quad (2.3)$$

The corresponding expanded term of the original derivative is

$$c_n(k, j, \ell) x^{x^x + kx - j} (\log x)^\ell.$$

Therefore

$$a(n) = \#\{(k, j, \ell) : c_n(k, j, \ell) \neq 0\}. \quad (2.4)$$

In particular, a negative coefficient contributes one term, not minus one.

2.2 Why the representation is unambiguous

Suppose a finite linear combination of $u^k v^j z^\ell$, after substitution, vanishes for all sufficiently large positive x . Group it by k and choose the largest k having a nonzero Laurent-polynomial coefficient in x and $\log x$. Division by x^{kx} makes every smaller- k group exponentially small relative to that coefficient. Within the largest group, choose the smallest j , and then the largest ℓ with nonzero coefficient. Its asymptotic magnitude cannot be canceled by the remaining Laurent-logarithmic terms. This is a contradiction. Hence the displayed monomials are linearly independent in the required finite sense.

This argument is also a safeguard against a representation-dependent term count: we are counting a specified canonical basis, not the summands in an arbitrarily factored expression.

2.3 The recurrence and its proof

Proposition 2.1 (Differential recurrence). *Starting from $P_0 = 1$, the polynomials satisfy*

$$P_{n+1} = u(z^2 + z + v)P_n + u(z + 1)\partial_u P_n - v^2\partial_v P_n + v\partial_z P_n. \quad (2.5)$$

Proof. Differentiate (2.2), apply the product rule, and substitute (2.1). The result is exactly (2.5). The base case is $f^{(0)} = f$. Induction proves the identity at every order, and also proves integrality of every coefficient. $\square$

For one coefficient $c = c_n(k, j, \ell)$ the seven contributions are as follows. All contributions with the same destination must be added before zero coefficients are removed.

Destination at order $n + 1$	Contribution	Source in the derivative
$(k + 1, j, \ell + 2)$	c	$f'/f: uz^2$
$(k + 1, j, \ell + 1)$	c	$f'/f: uz$
$(k + 1, j + 1, \ell)$	c	$f'/f: uv$
$(k, j, \ell + 1)$	kc	differentiation of u^k
(k, j, ℓ)	kc	differentiation of u^k
$(k, j + 1, \ell)$	$-jc$	differentiation of v^j
$(k, j + 1, \ell - 1)$	ℓc	differentiation of z^ℓ

Transitions with a zero multiplier are omitted. In particular, no negative logarithm exponent is ever introduced.

2.4 The first cancellation

The first two polynomials have the compact forms

$$P_1 = u(A + v), \quad (2.6)$$

$$P_2 = u^2(A + v)^2 + u((z + 1)A + (3z + 2)v - v^2). \quad (2.7)$$

Expansion gives $a(1) = 3$ and $a(2) = 12$. At the next order, the coefficient polynomial of uv^2 is

$$-(z + 1) + (\partial_z - 1)(3z + 2) = -4z.$$

Its constant term vanishes. The candidate envelope has 31 cells at order 3, but the derivative has 30 terms. A Boolean reachability calculation would fail at this very early example.

3 The support envelope and the cubic upper bound

Definition 3.1. For $n \geq 1$, $1 \leq k \leq n$, and $0 \leq j \leq n$, put

$$m_{n,k,j} = \max(0, k - j), \quad (3.1)$$

$$U_{n,k,j} = \begin{cases} n + k, & j = 0, \\ \min(n + k - j - 1, 2n - 2j), & j \geq 1. \end{cases} \quad (3.2)$$

Let $\mathcal{E}_n$ consist of the integer triples with these ranges and $m_{n,k,j} \leq \ell \leq U_{n,k,j}$.

Proposition 3.2 (Support containment). *For $n \geq 1$, $\text{supp } P_n \subseteq \mathcal{E}_n$.*

Proof. Consider a nonzero differentiation path, and let a, b, c, d, e, r, s count its uses of the seven transitions in the displayed table, in that order. Then

$$k = a + b + c, \quad j = c + r + s, \quad \ell = 2a + b + d - s, \quad n = a + b + c + d + e + r + s.$$

In particular,

$$\begin{aligned} n - j &= a + b + d + e, \\ \ell - (k - j) &= a + d + r \geq 0, \\ 2(n - j) - \ell &= b + d + 2e + s \geq 0, \\ n + k - j - \ell &= b + c + e + s \geq 0. \end{aligned}$$

If $j > 0$, at least one transition of type c or s must occur: the transition of type r has multiplier $-j$ and cannot create the first positive j . Thus $n + k - j - \ell \geq 1$ when $j > 0$. The remaining ranges follow immediately from the transition table and the first differentiation. Every coefficient is a sum over such paths, so cancellation can remove cells but cannot add a cell outside these constraints. $\square$

Proposition 3.3 (Size of the envelope). *For $n \geq 1$, $\#\mathcal{E}_n = B(n)$.*

Proof. For $j = 0$ there are $n + 1$ logarithm exponents for each n values of k , giving $n(n + 1)$ cells. For $j \geq 1$, set $d = n - j$, so $0 \leq d \leq n - 1$. The number of cells in this j -slice is

$$\sum_{k=1}^n (\min(d + k, 2d + 1) - \max(0, k - n + d)) = n(2d + 1) - d(d + 1).$$

Indeed, replacing each minimum by $2d + 1$ overcounts by $d(d + 1)/2$, and the sum of the subtracted maxima is another $d(d + 1)/2$. Consequently

$$\begin{aligned}\#\mathcal{E}_n &= n(n + 1) + \sum_{d=0}^{n-1} (n(2d + 1) - d(d + 1)) \\ &= n(n + 1) + n^3 - \frac{(n - 1)n(n + 1)}{3} \\ &= \frac{2n^3 + 3n^2 + 4n}{3}.\end{aligned}$$

□

This proves the upper bound in (1.2). More precisely, $\Delta(n)$ is exactly the number of zero coefficients *inside* $\mathcal{E}_n$. The polynomial $B(n)$ is not being guessed by interpolation: it is the exact count of a proved finite set of candidate monomials.

4 An independent exact formula for every coefficient

The recurrence is already an exact algorithm. An independent formula is valuable for two other reasons: it certifies possible cancellations after a modular computation, and it proves that every (k, j) row is nonzero.

4.1 Notation for Stirling numbers and auxiliary polynomials

We use signed Stirling numbers $s(n, k)$ of the first kind and Stirling numbers $S(n, k)$ of the second kind, with the conventions

$$y^n = \prod_{i=0}^{n-1} (y - i) = \sum_k s(n, k) y^k, \quad y^n = \sum_k S(n, k) y^k.$$

These conventions agree with DLMF §26.8 [4]. In particular, $s(n, k)$ has sign $(-1)^{n-k}$ and is nonzero for $1 \leq k \leq n$; $S(n, k) > 0$ on the same range.

For nonnegative d, p, q , define the noncentral Stirling numbers

$$T_d(p; q) = \sum_{r=0}^d \binom{d}{r} q^{d-r} S(r, p). \quad (4.1)$$

They can be computed without rational arithmetic:

$$T_0(p; q) = \mathbf{1}_{p=0}, \quad T_{d+1}(p; q) = (p + q)T_d(p; q) + T_d(p - 1; q). \quad (4.2)$$

Indices $p < 0$ or $p > d$ have value zero. The useful boundary values are

$$T_d(0; q) = q^d, \quad T_d(d; q) = 1, \quad T_d(p; 0) = S(d, p), \quad T_d(p; 1) = S(d + 1, p + 1),$$

where $0^0 = 1$. Equivalently,

$$e^{-w} \sum_{m \geq 0} (m + q)^d \frac{w^m}{m!} = \sum_{p=0}^d T_d(p; q) w^p. \quad (4.3)$$

To verify this identity, expand $(m + q)^d$, express powers of m in falling factorials, and sum $\sum_m m^p w^m / m! = w^p e^w$.

Define

$$Q_{j,r}(\beta) = \beta^j (\beta + 1)^{\bar{r}} = \prod_{i=0}^{j-1} (\beta - i) \prod_{i=1}^r (\beta + i). \quad (4.4)$$

These integer polynomials are easy to build by multiplying by linear factors. Notice that $\deg Q_{j,r} = j + r$.

4.2 A parameterized derivative identity

For parameters α, β , Taylor expansion with the increment xt gives

$$\frac{(x + xt)^{\alpha(x+xt)+\beta}}{x^{\alpha x+\beta}} = (1+t)^\beta \exp(\alpha x [tz + (1+t) \log(1+t)]). \quad (4.5)$$

Comparing coefficients of t^n and then of $(\alpha x)^d$ yields

$$\frac{d^n}{dx^n} x^{\alpha x+\beta} = x^{\alpha x+\beta-n} \sum_{d=0}^n (\alpha x)^d C_{n,d}(z, \beta), \quad (4.6)$$

where, with $j = n - d$,

$$C_{n,d}(z, \beta) = \sum_{s=0}^d \binom{n}{s} \frac{z^s}{(d-s)!} \partial_\beta^{d-s} Q_{j,d-s}(\beta). \quad (4.7)$$

For completeness, choose s factors tz from the d th power of the bracket in (4.5). The remaining factors contribute $(1+t)^{d-s} \log^{d-s}(1+t)$. Write the logarithm power as ∂_β^{d-s} acting on $(1+t)^{\beta+d-s}$ and use

$$[t^{n-s}](1+t)^{\beta+d-s} = \frac{(\beta+d-s)^{n-s}}{(n-s)!} = \frac{Q_{j,d-s}(\beta)}{(n-s)!}.$$

The factorial prefactors simplify to $\binom{n}{s}/(d-s)!$, proving the formula.

4.3 The coefficient oracle

Theorem 4.1 (Exact coefficient formula). *Suppose $(k, j, \ell) \in \mathcal{E}_n$, and put*

$$d = n - j, \quad h = d + k - \ell,$$

$$q_- = \max(0, k - \ell, k - d), \quad q_+ = \min(k, h, j).$$

Then

$$c_n(k, j, \ell) = \sum_{q=q_-}^{q_+} \binom{n}{\ell - k + q} \binom{h}{q} T_d(k - q; q) [\beta^h] Q_{j,h-q}(\beta). \quad (4.8)$$

An empty sum is zero. Outside the envelope, the coefficient is zero.

Proof. Expand the outer exponential:

$$f(x) = \sum_{m \geq 0} \frac{x^{mx} z^m}{m!}, \quad x^{mx} z^m = \partial_\beta^m x^{mx+\beta} \Big|_{\beta=0}.$$

Apply (4.6) with $\alpha = m$. For any polynomial $C(\beta)$,

$$\partial_\beta^m (e^{\beta z} C(\beta)) \Big|_{\beta=0} = \sum_{q=0}^m \frac{m!}{(m-q)!} z^{m-q} [\beta^q] C(\beta).$$

After division by $f = e^{uz}$, the sum over m occurring for fixed d, q is

$$\begin{aligned} e^{-uz} \sum_{m \geq q} \frac{m^d u^m z^{m-q}}{(m-q)!} &= u^q e^{-uz} \sum_{r \geq 0} (r+q)^d \frac{(uz)^r}{r!} \\ &= u^q \sum_{p=0}^d T_d(p; q) (uz)^p. \end{aligned}$$

To obtain $u^k z^\ell$, take $p = k - q$ and take $s = \ell - k + q$ from (4.7). Set $r = d - s = h - q$. Finally,

$$[\beta^q] \frac{\partial_\beta^r Q_{j,r}(\beta)}{r!} = \binom{q+r}{q} [\beta^{q+r}] Q_{j,r}(\beta),$$

and $q + r = h$. This proves (4.8); the stated bounds on q enforce $p \in [0, d]$, $s \in [0, d]$, and $q \leq j$.

The infinite series can be read as locally normally convergent analytic series for positive x near any fixed point, or as formal exponential identities followed by extraction of the finitely many relevant coefficients. $\square$

4.4 Three boundary checks

Useful independently checkable consequences are

$$[v^0]P_n = (z+1)^n \sum_{k=1}^n S(n, k) (uz)^k, \quad n \geq 1, \quad (4.9)$$

$$[u^n]P_n = (z(z+1) + v)^n, \quad (4.10)$$

$$[v^n]P_n = u^n. \quad (4.11)$$

The first follows either from (4.8) or from setting $v = 0$ in the recurrence. The second follows because reaching u -degree n requires every step to differentiate the outer factor f . For the third, the coefficient of the next highest v -power is obtained by multiplication by $u - n$, starting from 1.

5 A rigorous cubic lower bound

An upper bound alone does not establish cubic growth. We now show that a positive cubic number of the candidate cells must survive, without assuming that cancellations are rare.

5.1 Every row has its advertised highest coefficient

Let

$$R_{n,k,j}(z) = [u^k v^j] P_n(u, v, z).$$

Lemma 5.1 (Nonzero leading coefficient). *For every $1 \leq k \leq n$ and $0 \leq j \leq n$, the polynomial $R_{n,k,j}$ is nonzero and has degree exactly $U_{n,k,j}$.*

Proof. For $j = 0$, equation (4.9) gives

$$R_{n,k,0}(z) = S(n, k) z^k (z+1)^n.$$

Now let $j \geq 1$ and put $d = n - j$. If $k \leq d$, the proposed highest exponent is $\ell = d + k - 1$, so $h = 1$ in (4.8). Only $q = 0, 1$ contribute. The coefficient is

$$(-1)^{j-1}(j-1)! \left(\binom{n}{d-1} S(d, k) + \binom{n}{d} S(d+1, k) \right), \quad (5.1)$$

which is nonzero. Here we used $[\beta]Q_{j,0} = [\beta]Q_{j,1} = (-1)^{j-1}(j-1)!$.

If $k \geq d + 1$, the highest exponent is $\ell = 2d$, and $h = k - d$. The only possible summand is $q = h$, giving

$$[z^{2d}]R_{n,k,j}(z) = \binom{n}{d} s(j, k - d). \quad (5.2)$$

It is nonzero because $1 \leq k - d \leq j$. The support envelope already rules out higher exponents, completing the proof. $\square$

5.2 A high-multiplicity root

Lemma 5.2 (Divisibility). *The row polynomial is divisible by*

$$z^{\max(k-j,0)}(z+1)^{M_{n,k,j}}, \quad M_{n,k,j} = \max(n-2j, k-j, 0). \quad (5.3)$$

Proof. Extracting $u^k v^j$ from (2.5) gives

$$\begin{aligned} R_{n+1,k,j} &= z(z+1)R_{n,k-1,j} + R_{n,k-1,j-1} + k(z+1)R_{n,k,j} \\ &\quad + (\partial_z - (j-1))R_{n,k,j-1}. \end{aligned} \quad (5.4)$$

Rows with out-of-range indices are zero. Begin with $R_{0,0,0} = 1$. For the factor $z+1$, the first and third terms increase the inherited multiplicity by one. The second term has inherited multiplicity at least $\max(n-2j+2, k-j, 0)$. The derivative in the last term can reduce its inherited multiplicity by at most one; before differentiation it is at least $\max(n-2j+2, k-j+1, 0)$. Each is sufficient for the target exponent $\max(n+1-2j, k-j, 0)$. This proves the $(z+1)$ assertion by induction. The same argument with the factor z proves the exponent $\max(k-j, 0)$. Since z and $z+1$ are relatively prime, both divisibilities hold together. $\square$

Lemma 5.3 (Multiplicity forces many terms). *A nonzero polynomial with a nonzero root of multiplicity at least M has at least $M+1$ nonzero coefficients.*

Proof. Suppose $p(z) = \sum_{i=1}^t b_i z^{e_i}$, with distinct nonnegative integers e_i and $b_i \neq 0$. If a root $\alpha \neq 0$ has multiplicity at least t , then $(z\partial_z)^r p(\alpha) = 0$ for $0 \leq r < t$. Thus

$$\sum_{i=1}^t e_i^r b_i \alpha^{e_i} = 0, \quad 0 \leq r < t.$$

The matrix (e_i^r) is an invertible Vandermonde matrix, contradicting $b_i \alpha^{e_i} \neq 0$. Therefore the multiplicity is at most $t-1$. $\square$

5.3 Summing the mandatory terms

By Lemmas 5.1–5.3, each (k, j) row contributes at least $1 + M_{n,k,j}$ nonzero monomials. Rows are disjoint in the canonical basis, so

$$a(n) \geq \sum_{k=1}^n \sum_{j=0}^n (1 + \max(n-2j, k-j, 0)). \quad (5.5)$$

For fixed j , the sum of the maxima over k is

$$\sum_{k=1}^n M_{n,k,j} = \begin{cases} n(n-2j) + \frac{j(j+1)}{2}, & 0 \leq j \leq \lfloor n/2 \rfloor, \\ \frac{(n-j)(n-j+1)}{2}, & j > \lfloor n/2 \rfloor. \end{cases} \quad (5.6)$$

In the first case, start with $n-2j$ in each row and add the excess $1 + \dots + j$. In the second case, only $k > j$ contributes. Summing these elementary progressions, and adding the $n(n+1)$ ones in (5.5), gives exactly $L(n)$ in (1.2). This completes the proof of the growth bounds in Theorem 1.1.

In particular,

$$\frac{7}{24} \leq \liminf_{n \rightarrow \infty} \frac{a(n)}{n^3} \leq \limsup_{n \rightarrow \infty} \frac{a(n)}{n^3} \leq \frac{2}{3}. \quad (5.7)$$

These constants are proved bounds, not estimates of an experimentally fitted leading coefficient.

6 Exact cancellation families

6.1 Logarithm-free coefficients and derivatives of a falling factorial

Let $F_n(y) = y^n$. In (4.8), setting $\ell = 0$ forces $q = k$, so for $d = n - j$ and $1 \leq k \leq j$,

$$\boxed{c_n(k, j, 0) = \frac{k^d}{d! k!} F_n^{(d+k)}(d)}. \quad (6.1)$$

Although written using factorial quotients, this expression is an integer; the implementation instead uses integer polynomial coefficients and binomial coefficients. Formula (6.1) reduces every logarithm-free cancellation to an exact derivative evaluation of a single elementary polynomial.

Proposition 6.1 (Central cancellations at every odd order). *For odd $n \geq 1$,*

$$\Delta(n) \geq \left\lfloor \frac{n+1}{4} \right\rfloor. \quad (6.2)$$

Proof. Write $n = 2d + 1$ and set $j = d + 1$. Then

$$F_n(d + \beta) = \beta \prod_{r=1}^d (\beta^2 - r^2)$$

is odd. For each $1 \leq k \leq d + 1$ with $k \equiv d \pmod{2}$, its coefficient of β^{d+k} vanishes. Formula (6.1) gives a missing monomial $(k, d + 1, 0)$. There are $\lfloor (d + 1)/2 \rfloor = \lfloor (n + 1)/4 \rfloor$ such values of k . $\square$

Thus $a(n) = B(n)$ is impossible at every odd $n \geq 3$. This does not resolve whether equality occurs infinitely often at even orders. Nor does it classify all logarithm-free cancellations: additional integer derivative zeros occur away from the central symmetry.

6.2 An infinite family with positive logarithm exponent

It is tempting to guess that all cancellations are logarithm-free. The following family disproves that guess.

Since $f = e^g$ with $g = uz$, the u^{n-1} part of $f^{(n)}/f$ consists of the partition type having one block of size two and all other blocks of size one. Equivalently, induction in (2.5) gives

$$[u^{n-1}]P_n = \binom{n}{2}(A+v)^{n-2}((z+1)A + (3z+2)v - v^2), \quad n \geq 2. \quad (6.3)$$

For $2 \leq j \leq n-1$, put $m = n - j - 1$ and

$$a = \binom{n-2}{j}, \quad b = \binom{n-2}{j-1}, \quad c = \binom{n-2}{j-2}.$$

The coefficient of v^j , after removing the common factor $\binom{n}{2}$, is

$$z^m(z+1)^m \left[-cz^2 + (a+3b-c)z + (a+2b) \right]. \quad (6.4)$$

Its coefficient of z^{2m+1} is

$$a + 3b - (m+1)c. \quad (6.5)$$

Using

$$\frac{a}{c} = \frac{m(m+1)}{j(j-1)}, \quad \frac{b}{c} = \frac{m+1}{j-1},$$

this vanishes exactly when $m = j^2 - 4j$ in the stated range.

Proposition 6.2 (Non-logarithm-free cancellation family). *For every integer $t \geq 2$, let*

$$n = t^2 + t - 1, \quad k = n - 1, \quad j = t + 2, \quad \ell = 2t^2 - 7. \quad (6.6)$$

Then $(k, j, \ell) \in \mathcal{E}_n$, $\ell > 0$, and $c_n(k, j, \ell) = 0$.

Proof. Substitute $j = t + 2$ into $m = j^2 - 4j$: it gives $m = t^2 - 4$ and $n = m + j + 1 = t^2 + t - 1$. Then $\ell = 2m + 1$ is positive, is inside the envelope, and its coefficient vanishes by (6.5). The next higher coefficient is $-\binom{n}{2}c \neq 0$, so this is a genuine internal missing term. $\square$

The first corresponding orders are 5, 11, 19, 29, 41, 55, $\dots$, with missing triples

$$(4, 4, 1), (10, 5, 11), (18, 6, 25), (28, 7, 43), \dots$$

These are additional to the logarithm-free central cancellations. Since all orders in (6.6) are odd, this proves $\Delta(n) \geq \lfloor (n+1)/4 \rfloor + 1$ along that infinite subsequence.

7 Algorithms and complexity

7.1 The dependency-free exact recurrence

The most direct implementation stores a dictionary from triples to integer coefficients. At each differentiation it applies the seven transitions, combines equal keys, and removes zeros. A short reference version is given in Appendix A. It uses no symbolic-algebra system and no numerical approximation.

The supplied production version packs a triple into one integer key. For a requested maximum order N , take

$$b = N + 2, \quad c = 2N + 3, \quad \text{key}(k, j, \ell) = (kb + j)c + \ell.$$

Incrementing k, j, ℓ corresponds to adding $bc, c, 1$, respectively. The chosen bases exceed every possible exponent, so no carries can identify different monomials. The packed representation reduces tuple allocation and dictionary overhead; it does not change the mathematics.

Proposition 7.1 (Arithmetic complexity). *The exact recurrence computes the entire prefix through N in $O(N^4)$ integer arithmetic operations and $O(N^3)$ live integer coefficients. Each coefficient has $O(N \log N)$ bits.*

Proof. At order n there are at most $B(n) = O(n^3)$ terms, and each contributes to at most seven destinations. Summing through N gives $O(N^4)$ operations. Only the current and next dictionaries are needed, giving $O(N^3)$ storage.

Let H_n be the sum of the absolute values of the coefficients of P_n . The absolute sum of all multipliers leaving a term is $3 + 2k + j + \ell \leq 3 + 5n$. Therefore

$$H_{n+1} \leq (5n + 3)H_n, \quad H_0 = 1, \quad H_n \leq \prod_{r=0}^{n-1} (5r + 3) \leq 5^n n!.$$

Every individual coefficient satisfies this bound, so its bit length is $O(n \log n)$. □

This is a unit-cost *integer-arithmetic* bound, not a claim that arbitrarily large integers are machine words. With elementary small-integer multiplication, a conservative bit bound is $O(N^5 \log^2 N)$ time and $O(N^4 \log N)$ coefficient-storage bits. Hashing and object overhead affect practical memory. We do not claim the algorithm is asymptotically optimal.

7.2 A deterministic modular sieve with exact refinement

Most coefficients need never be constructed as large integers. Fix any integer modulus $M \geq 2$, and run the same recurrence in $\mathbb{Z}/M\mathbb{Z}$. If a residue is nonzero, the original integer coefficient is certainly nonzero. Only zero residues need further attention.

The exact single-term algorithm is:

1. Compute $P_n \bmod M$ by the seven-transition recurrence.
2. Enumerate the triples in $\mathcal{E}_n$ whose residues are zero.
3. Evaluate each such coefficient with (4.8), using integer arithmetic, and count only the coefficients that are actually zero.
4. Return $B(n)$ minus that exact zero count.

A zero residue is *never* accepted as proof of a zero coefficient. Consequently this algorithm is correct for every admissible modulus, including small or composite moduli. No randomness, primality assumption, or conjecture about support is required. Tests deliberately use moduli 2, 7, and 12 to force many false modular zeros.

The supplied NumPy implementation uses signed 64-bit arrays and the modulus 1 000 000 007 by default. At order m , the array has shape $(m + 1, m + 1, 2m + 1)$. The next array has shape

$(m + 2, m + 2, 2m + 3)$, and each transition becomes a shifted slice addition. All entries are reduced modulo M after each step. The implementation checks

$$(10n + 10)M < 2^{63}$$

before starting; this conservative condition bounds the intermediate products and sums before modular reduction. Overflow would invalidate the certificate, so the check is part of correctness rather than merely an optimization detail.

The dense modular pass takes $O(n^4)$ fixed-width operations and $O(n^3)$ fixed-width words. Let $Z_M(n)$ denote the number of zero residues inside the envelope. The oracle tables $Q_{j,r}$ and $T_d(p; q)$, and Pascal's triangle for binomial coefficients, are built lazily. For a single target n , filling all potentially needed tables takes $O(n^3)$ integer arithmetic operations and storage; each candidate coefficient then requires at most $O(n)$ integer products and additions. Thus a conservative oracle bound is

$$O(n^3 + nZ_M(n)) \subseteq O(n^4).$$

In practice the lazy tables are much smaller when the modular pass leaves few candidates. Unlike the reference recurrence, the oracle may multiply two large integers, so its unit-arithmetic bound should again not be read as a bit-operation bound.

7.3 Which implementation to use

For a whole prefix, the dependency-free recurrence has the simplest predictable cost and automatically supplies every intermediate count. For an isolated larger index, the modular method usually avoids nearly all large-integer coefficient work. Running a new modular pass separately for every prefix index would discard this advantage and incur repeated work.

Neither method differentiates a transcendental expression in a computer algebra system. Both return the exact OEIS term count, including signed cancellations, rather than an expression-size estimate.

8 Computational validation and results

8.1 Checks actually performed

The accompanying files record ten passing test groups. The tests compare the reference recurrence with all 43 published A281434 values through $n = 42$, and compare the generated prefix with all 80 published A352697 deficits through $n = 80$ [1, 2]. Every candidate coefficient through $n = 12$ is checked against the independent formula (4.8). The two methods are also compared through $n = 12$ for each of four moduli, including the deliberately small and composite choices described above.

Additional tests verify the envelope, its exact cardinality, the lower-bound sum, the $(z + 1)$ multiplicities, the two leading-coefficient formulas, the odd central cancellations, and initial instances of the infinite non-logarithm-free family. Such finite tests check the implementations and help detect algebraic mistakes; the universal statements are justified by the proofs, not by the tests.

The integer recurrence was run through $n = 100$. The modular implementation was separately checked against it at $n = 20, 40, 60, 80, 100$. Selected larger computations used the modular pass and exact verification of all candidate zeros. The archive contains the complete generated prefix, the tests, and the selected missing triples.

8.2 Selected exact values

n	$a(n)$	$B(n)$	$\Delta(n)$
0	1	–	–
1	3	3	0
2	12	12	0
3	30	31	1
5	113	115	2
11	1 017	1 023	6
20	5 760	5 760	0
40	44 320	44 320	0
60	147 680	147 680	0
80	347 838	347 840	2
100	676 800	676 800	0
101	697 178	697 203	25
150	2 272 700	2 272 700	0
151	2 318 265	2 318 303	38
200	5 373 600	5 373 600	0

For example, the two missing triples at $n = 80$ are $(33, 36, 0)$ and $(42, 45, 0)$. At $n = 101$, all 25 missing terms are supplied by the central logarithm-free symmetry in Proposition 6.1. These are exact statements about the recorded finite computations.

8.3 Measured performance

On the execution environment used for this report, the packed integer recurrence generated the full prefix $0 \leq n \leq 100$ in approximately 26.16 seconds. Selected single-index modular runs were:

n	Modular zero candidates	Actual zeros	Elapsed seconds
40	0	0	0.021
60	0	0	0.096
80	2	2	0.342
100	0	0	0.820
101	25	25	0.698
150	0	0	4.970
151	38	38	5.460
200	0	0	17.769

These are individual measurements, not a controlled benchmark suite or a promise of timings on other hardware. In particular, computing one target by the modular method and computing an entire prefix by the integer method are different workloads. The exact timings, interpreter information, and modulus are saved in `data/benchmarks.csv` and `data/environment.json`.

The fact that every zero residue in this small selection was a genuine zero is not assumed by the algorithm. The small-modulus tests exercise the opposite situation and confirm that false modular zeros are rejected by the exact oracle.

9 What is and is not established asymptotically

The unconditional conclusion is cubic-order growth, with the fully explicit bounds (1.2). Consequently, $a(n) = n^{3+o(1)}$ in the logarithmic-exponent sense. The lower bound remains valid even if many more cancellations occur at orders far beyond the computed range.

There are three progressively stronger claims:

$$\begin{array}{ll} \text{Proved here:} & a(n) = \Theta(n^3). \\ \text{Not proved here:} & a(n) \sim \frac{2}{3}n^3. \\ \text{Stronger OEIS conjecture:} & a(n) = \frac{2}{3}n^3 + n^2 + O(n). \end{array}$$

The last statement is equivalent to $\Delta(n) = O(n)$, given the exact formula for $B(n)$, and implies the middle statement. Neither follows from agreement with $B(n)$ at many computed indices.

A proof of the middle statement needs only a subcubic bound on the number of holes; it need not classify every cancellation. In particular, the entire plane $\ell = 0$ has only $O(n^2)$ candidate cells. Thus even a complete failure to classify the logarithm-free derivative zeros would not by itself prevent proving the leading equivalent. The remaining issue is a uniform bound on cancellations with $\ell > 0$ across the three-dimensional envelope. Formula (4.8) supplies an explicit integer expression for those coefficients, and the family in Proposition 6.2 shows why a blanket nonvanishing assertion would be false.

For the stronger linear-deficit conjecture, even the logarithm-free plane requires more delicate control. Its central symmetry already forces $\Delta(n)$ to be at least a positive linear function along the odd orders, so an $o(n)$ bound is impossible. The appropriate target is an upper bound of matching order, not an assertion that the derivative eventually fills its whole envelope.

The computation problem is therefore resolved by unconditional exact algorithms, and the polynomial growth exponent is established. Determining the exact leading asymptotic constant remains a separate nonvanishing problem in this analysis.

A A compact exact implementation

The following version emphasizes transparency. The archive's `code/a281434.py` additionally provides packed keys, the coefficient oracle, modular arrays, input validation, and a command-line interface.

```
from collections import defaultdict

def a281434_prefix(N):
    if not isinstance(N, int) or isinstance(N, bool) or N < 0:
        raise ValueError("N must be a nonnegative integer")
    p = {(0, 0, 0): 1}
    values = [1]
    for _ in range(N):
        q = defaultdict(int)
        for (k, j, ell), c in p.items():
            q[k+1, j, ell+2] += c
            q[k+1, j, ell+1] += c
            q[k+1, j+1, ell] += c
            if k:
                q[k, j, ell+1] += k*c
                q[k, j, ell] += k*c
            if j:
                q[k, j+1, ell] -= j*c
            if ell:
                q[k, j+1, ell-1] += ell*c
        p = {key: c for key, c in q.items() if c}
        values.append(len(p))
    return values
```

The only potentially negative transition is the one with multiplier $-j$. Deleting it, taking absolute values before accumulation, or replacing the dictionary by a set changes the sequence.

B Reproducing the supplied computations

Python 3.9 or later is sufficient for the dependency-free method. The modular method additionally requires NumPy. From the archive's `code` directory, the commands below produce a single term, inspect missing monomials, create a prefix, and run the tests.

```
python a281434.py 100 --method modular
python a281434.py 80 --method modular --holes
python a281434.py 100 --prefix
python -m unittest -v test_a281434.py
python run_experiments.py --reference
python run_experiments.py 20 40 60 80 100 101 150 151 200
```

The first command returns 676800. The second returns 347838 and the two missing triples listed in the article. The full prefix and related deficits are in `data/b281434.txt` and `data/b352697.txt`. These are generated data files accompanying this report, not claims that the OEIS has incorporated the additional terms.

The LaTeX source builds with

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Repeated runs resolve the table of contents and cross-references. The bundled `build.sh` and `build.ps1` perform these commands.

References

- [1] Vladimir Reshetnikov, *A281434: Number of terms in the fully expanded n -th derivative of $x^{(x^x)}$* , The On-Line Encyclopedia of Integer Sequences, entry created October 5, 2017. <https://oeis.org/search?q=id:A281434&fmt=text>. Accessed September 20, 2026.
- [2] Vladimir Reshetnikov, *A352697: $a(n) = A037237(n - 1) - A281434(n)$* , The On-Line Encyclopedia of Integer Sequences, March 29, 2022. <https://oeis.org/A352697>. Accessed September 20, 2026.
- [3] N. J. A. Sloane and contributors, *A037237: Expansion of $(3 + x^2)/(1 - x)^4$* , The On-Line Encyclopedia of Integer Sequences. <https://oeis.org/A037237>. Accessed September 20, 2026.
- [4] NIST Digital Library of Mathematical Functions, §26.8: *Set Partitions—Stirling Numbers*. <https://dlmf.nist.gov/26.8>. Accessed September 20, 2026.